

CAP Theorem - Availability

1. What is Availability in CAP?

Availability means:

None

Every request receives a response

Even if some nodes fail, the distributed system remains operational.

The response may be:

- Correct latest data
- Older cached data
- Replica data
- Temporary fallback response

But the system should respond.

2. Simple Meaning

If 3 database nodes exist and 1 or 2 fail:

None

System still serves users

No complete outage.

3. Example

E-commerce product service:

Even if one region fails, users should still be able to:

- Search products
- View items
- Add to cart

Maybe inventory count is slightly stale, but service works.

4. Real-World Analogy

ATM network:

If one ATM server fails, users can still withdraw from another ATM.

That shows availability.

5. Key Rule in CAP

Availability does **not** guarantee latest data.

Example:

None

Price updated ₹999 → ₹899

Some node may still return:

None

₹999

But request succeeded.

6. Why Availability Matters

Critical for:

Social media

Search engines

Messaging apps

E-commerce browsing

News sites

Streaming platforms

Downtime causes:

- Revenue loss
- Bad user experience
- Reputation damage

7. How Systems Achieve Availability

A. Replication

Multiple copies of data.

If one node fails, another serves traffic.

B. Load Balancers

Traffic routed to healthy nodes.

C. Auto Failover

Dead leader replaced automatically.

D. Retry + Timeout Handling

Temporary network issues tolerated.

E. Caching / CDN

Serve cached responses.

Examples:

- Amazon Web Services load balancing
- Redis cache
- Cloudflare CDN

8. Availability During Partition

Suppose Node A cannot talk to Node B.

To remain available:

None

Both sides continue serving requests

But data may diverge temporarily.

This sacrifices consistency.

9. Example: AP System

Two nodes:

None

Node A = old balance ₹1000

Node B = new balance ₹1200

Reads from either node still return response.

That is high availability.

10. Availability vs Reliability

Term	Meaning
Availability	System accessible now
Reliability	Works correctly over time

11. Availability vs Consistency

Property	Meaning
Consistency	Latest same data everywhere
Availability	Every request gets response

Sometimes cannot guarantee both during partition.

12. Real Systems Favoring Availability

Examples often designed toward availability/eventual consistency:

- Apache Cassandra
- Riak
- Amazon DynamoDB (configurable behaviors)

13. HLD Design Use Cases

Choose availability-first for:

None

Product catalog

Feed systems

Likes count

Comments

Search autocomplete

Recommendations

Users prefer response over perfect freshness.

14. Interview Answer Example

Q: What is availability in CAP theorem?

Availability means every request receives a non-error response, even when some nodes fail or become unreachable, though the response may not always contain the latest data.

15. Short Revision Notes

None

Availability:

Every request gets response

Even if nodes fail

May return stale data

Achieved via:

replication

failover

load balancing

cache

16. One-Line Summary

Availability in CAP means the system continues responding to all requests despite failures, even if some responses may be stale or inconsistent temporarily.